

Tutorial 8 — Exercise 1: Deriving the F -Test (Solutions)

Introductory Econometrics — TA Session

Duration: 50 minutes | Based on Lecture 14 (Marmer)

Consider the log-wage equation from Lecture 14:

$$\ln(\text{Wage}_i) = \beta_0 + \beta_1 \text{Exp}_i + \beta_2 \text{Exp}_i^2 + \beta_3 \text{PrevExp}_i + \beta_4 \text{PrevExp}_i^2 + \beta_5 \text{Educ}_i + U_i$$

where Exp is current job experience, PrevExp is previous experience, and Educ is years of education ($k = 5$ regressors plus intercept). We want to test whether previous experience has *any* effect on wages. This requires testing β_3 and β_4 jointly. You will derive the test statistic that does this correctly.

Step 1. Hypotheses. The null imposes $q = 2$ restrictions simultaneously:

$$H_0: \beta_3 = 0 \text{ and } \beta_4 = 0 \quad \text{against} \quad H_1: \beta_3 \neq 0 \text{ or } \beta_4 \neq 0.$$

(a) Suppose H_0 is true. You run two separate size- α t -tests and reject whenever *at least one* rejects. Using $P(A \cup B) = P(A) + P(B) - P(A \cap B)$, show that the false rejection probability exceeds α .

Solution

We want to show that using two separate t -tests inflates the Type I error rate — that is, the probability of falsely rejecting H_0 when it is true becomes larger than the nominal level α .

Step 1: Define the events. Let:

- $A = \{|T_3| > t_{n-k-1, 1-\alpha/2}\}$: the event that the first t -test rejects $\beta_3 = 0$.
- $B = \{|T_4| > t_{n-k-1, 1-\alpha/2}\}$: the event that the second t -test rejects $\beta_4 = 0$.

Step 2: Determine the individual rejection probabilities. Under H_0 , both $\beta_3 = 0$ and $\beta_4 = 0$ are true. Each t -test is designed so that, when its individual null is true, it rejects with probability exactly α . Therefore:

$$P(A) = \alpha, \quad P(B) = \alpha.$$

Step 3: Write the probability of the combined procedure. Our combined procedure rejects H_0 whenever *at least one* of the two t -tests rejects. That is, we reject when event $A \cup B$ (“ A or B ”) occurs. By the inclusion–exclusion formula from probability theory:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B).$$

Substituting $P(A) = P(B) = \alpha$:

$$P(A \cup B) = \alpha + \alpha - P(A \cap B) = 2\alpha - P(A \cap B).$$

Step 4: Bound $P(A \cap B)$. The term $P(A \cap B)$ is the probability that *both* tests reject simultaneously. We do not know this probability exactly (it depends on the correlation between T_3 and T_4), but we can bound it. Since $A \cap B \subseteq A$ (if both reject, then certainly the first one rejects):

$$P(A \cap B) \leq P(A) = \alpha.$$

Step 5: Combine. Substituting this upper bound into our expression:

$$P(A \cup B) = 2\alpha - P(A \cap B) \geq 2\alpha - \alpha = \alpha.$$

This proves that the false rejection probability is *at least* α . The combined procedure over-rejects.

How bad can it get? In the extreme case where T_3 and T_4 are independent (which happens when $PrevExp$ and $PrevExp^2$ are uncorrelated after partialling out the other regressors), $P(A \cap B) = P(A) \cdot P(B) = \alpha^2$. Then:

$$P(A \cup B) = 2\alpha - \alpha^2 = 2(0.05) - (0.05)^2 = 0.10 - 0.0025 = 0.0975 \approx 10\%.$$

So a “5% test” actually rejects under H_0 almost 10% of the time — roughly double the intended rate.

The lesson: stacking individual t -tests inflates the Type I error. With q separate tests, the false rejection rate can reach $q\alpha$. We need a *single* test statistic that handles all q restrictions simultaneously and keeps the Type I error at exactly α . That statistic is the F -statistic.

Step 2. Unrestricted and restricted models. We compare the fit of two models:

- **Unrestricted:** all regressors included $\rightarrow$ compute $SSR_{ur} = \sum \hat{U}_i^2$. Has $k + 1 = 6$ parameters; residual df $= n - k - 1$.
- **Restricted** (impose H_0 : drop $PrevExp$, $PrevExp^2$): $\ln(Wage_i) = \beta_0 + \beta_1 Exp_i + \beta_2 Exp_i^2 + \beta_5 Educ_i + U_i$ $\rightarrow$ compute SSR_r . Has 4 parameters.

(b) Verify that the parameter difference equals q . Explain why $SSR_r \geq SSR_{ur}$ always.

Solution

Parameter count:

- **Unrestricted model:** 6 free parameters: β_0 (intercept), β_1 (Exp), β_2 (Exp^2), β_3 ($PrevExp$), β_4 ($PrevExp^2$), β_5 ($Educ$).
- **Restricted model:** 4 free parameters: β_0 (intercept), β_1 (Exp), β_2 (Exp^2), β_5 ($Educ$). We lost β_3 and β_4 because H_0 forces them to be zero.
- **Difference:** $6 - 4 = 2 = q$. Each restriction removes exactly one free parameter from the model.

Why $SSR_r \geq SSR_{ur}$ always:

OLS chooses coefficients to minimise the sum of squared residuals. Think of it as an optimisation problem:

- The **unrestricted** model minimises SSR over *all possible values* of $(\beta_0, \beta_1, \beta_2, \beta_3, \beta_4, \beta_5)$. It is free to pick any combination.
- The **restricted** model minimises SSR over a *smaller set*: only values where $\beta_3 = 0$ and $\beta_4 = 0$. It has fewer levers to pull.

The restricted model’s feasible set is a *subset* of the unrestricted model’s feasible set. (Everything the restricted model can do, the unrestricted model can also do — and more.) Therefore, the minimum SSR over the smaller set cannot be lower than the minimum over the larger set:

$$SSR_r \geq SSR_{ur} \quad \text{always.}$$

Intuition: giving OLS more freedom (more parameters to optimise) can only help the fit or leave it unchanged. It can never make it worse. If the true β_3 and β_4 really are zero, then the unrestricted model will estimate them close to zero anyway, and $SSR_r \approx SSR_{ur}$. If they are not zero, forcing them to be zero will hurt the fit, and $SSR_r > SSR_{ur}$.

Step 3. The F -statistic. If H_0 is true, $SSR_r - SSR_{ur}$ should be small (restrictions barely hurt fit). If false, it should be large. The F -statistic formalises this:

$$F = \frac{(SSR_r - SSR_{ur}) / q}{SSR_{ur} / (n - k - 1)}$$

(c) Interpret the numerator and denominator. Why divide by q ? What does $SSR_{ur} / (n - k - 1)$ estimate?

Solution

The F -statistic is a ratio with a clear economic interpretation. Let us break it down piece by piece.

Numerator: $(SSR_r - SSR_{ur})/q$

- $SSR_r - SSR_{ur}$ is the **total increase in the residual sum of squares** when we impose H_0 . It measures the total amount of explanatory power lost by dropping $PrevExp$ and $PrevExp^2$ from the model.
- We divide by q (the number of restrictions) to get the **average increase per restriction**. This normalisation is essential: a test with $q = 10$ restrictions will naturally produce a larger total gap $SSR_r - SSR_{ur}$ than a test with $q = 2$, even if each individual restriction is equally innocuous. Dividing by q puts all F -tests on the same scale, regardless of how many restrictions are being tested.
- In our example: $q = 2$, so the numerator is the average fit loss per excluded variable (per restriction on previous experience).

Denominator: $SSR_{ur}/(n - k - 1)$

- This is the **mean squared error** (MSE) from the unrestricted model. Recall from earlier in the course that:

$$\hat{\sigma}^2 = \frac{SSR_{ur}}{n - k - 1}$$

is an unbiased and consistent estimator of σ^2 , the variance of the error term U_i .

- Think of it as the **background noise level**: how much residual variation is “normal” for this model. Even if all your regressors are correct, there will always be some residual variation due to unobserved factors captured by U_i . The denominator measures the typical size of this variation.
- We use the *unrestricted* model’s MSE (not the restricted one’s) because the unrestricted model provides the best available estimate of σ^2 — it does not impose potentially false restrictions.

Putting it together: F as a signal-to-noise ratio.

The F -statistic asks a precise question: “Is the fit loss from imposing H_0 (per restriction) large relative to the background noise?”

- If F is **close to zero**: the restrictions barely hurt the fit. The gap $SSR_r - SSR_{ur}$ is small compared to the noise level. This is consistent with H_0 being true — the excluded variables were not doing much.
- If F is **large**: the restrictions cause a substantial deterioration in fit that is hard to explain by sampling variation alone. This is evidence against H_0 — the excluded variables were genuinely important.

(d) Compute F using Lecture 14 data: $SSR_{ur} = 96.998$, $SSR_r = 99.463$, $n = 526$, $k = 5$, $q = 2$.

Solution

We substitute each value into the formula, computing every intermediate step.

Step 1: Compute the numerator.

First, the total fit cost of imposing H_0 :

$$SSR_r - SSR_{ur} = 99.463 - 96.998 = 2.465.$$

This tells us that dropping $PrevExp$ and $PrevExp^2$ increased the sum of squared residuals by 2.465. Now divide by the number of restrictions to get the average cost per restriction:

$$\frac{SSR_r - SSR_{ur}}{q} = \frac{2.465}{2} = 1.2325.$$

On average, each restriction increases the SSR by 1.2325.

Step 2: Compute the denominator.

First, the residual degrees of freedom in the unrestricted model:

$$n - k - 1 = 526 - 5 - 1 = 520.$$

Now the mean squared error:

$$\frac{SSR_{ur}}{n - k - 1} = \frac{96.998}{520} = 0.18654.$$

This is $\hat{\sigma}^2 = 0.187$, the estimated variance of the error term in the unrestricted model.

Step 3: Compute F .

$$F = \frac{1.2325}{0.18654} \approx 6.61.$$

Interpretation: the fit loss per restriction (1.23) is about 6.6 times larger than the background noise level (0.19). This is a substantial ratio — it suggests that dropping *PrevExp* and *PrevExp*² hurts the model fit by more than we would expect from sampling variation alone. Whether 6.61 is “large enough” to reject H_0 depends on the critical value, which we determine in Step 5.

Step 4. Distribution and degrees of freedom. Under H_0 and normality ($U_i|X \sim \mathcal{N}(0, \sigma^2)$): $F \sim F_{q, n-k-1}$. This follows because $(SSR_r - SSR_{ur})/\sigma^2 \sim \chi_q^2$ and $SSR_{ur}/\sigma^2 \sim \chi_{n-k-1}^2$, independently, and the F -distribution is defined as the ratio of two independent χ^2 's divided by their df (with σ^2 cancelling).

(e) State the numerator and denominator df for the wage example. Explain why $F \geq 0$ always.

Solution

Degrees of freedom:

The $F_{q, n-k-1}$ distribution has two parameters:

1. **Numerator df** = $q = 2$. This is the number of restrictions being tested. It comes from the χ_q^2 distribution in the numerator of F . In our wage example, we are testing two restrictions ($\beta_3 = 0$ and $\beta_4 = 0$), so $q = 2$.
2. **Denominator df** = $n - k - 1 = 526 - 5 - 1 = 520$. This is the residual degrees of freedom from the **unrestricted** model. It comes from the χ_{n-k-1}^2 distribution in the denominator. Here, $k = 5$ is the number of regressors in the *unrestricted* model.

So in this example: $F \sim F_{2, 520}$ under H_0 .

Common mistake: students sometimes use the residual df from the *restricted* model ($n - 3 - 1 = 522$) instead of the unrestricted model (520). The denominator df *always* comes from the unrestricted model, because the denominator of the F -statistic uses SSR_{ur} and its degrees of freedom $n - k - 1$.

Why $F \geq 0$ always:

The F -statistic is a ratio. Consider each part separately:

- **Numerator:** $SSR_r - SSR_{ur} \geq 0$ because imposing restrictions can never improve the OLS fit (as we showed in part (b)). Dividing by $q > 0$ preserves the sign.
- **Denominator:** $SSR_{ur} > 0$ because a perfect fit (every residual exactly zero) is essentially impossible with real data, and $n - k - 1 > 0$ as long as we have more observations than parameters.

A non-negative number divided by a strictly positive number is non-negative: $F \geq 0$.

What does the value of F tell us?

- $F = 0$: the restrictions did not hurt the fit at all — $SSR_r = SSR_{ur}$. This is the strongest possible evidence *in favour* of H_0 .
- F close to zero: the restrictions barely hurt the fit. Consistent with H_0 .
- F large: the restrictions cause a substantial deterioration. Evidence against H_0 .

Step 5. Decision rule. At significance level α : reject H_0 when $F > F_{q, n-k-1, 1-\alpha}$.

The F -test is always one-sided (reject for large F only) and the denominator df comes from the **unrestricted** model.

(f) The critical value is $F_{2,520,0.95} \approx 3.01$ and the p -value ≈ 0.0015 . Do you reject? Interpret.

Solution

Decision using the critical value:

We compare the observed F -statistic to the critical value of the $F_{2,520}$ distribution at $\alpha = 0.05$:

$$F_{\text{obs}} = 6.61 \quad \text{vs.} \quad F_{2,520,0.95} = 3.01.$$

Since $6.61 > 3.01$, the observed F -statistic falls in the rejection region. We **reject** H_0 at the 5% significance level.

Decision using the p -value:

The p -value is $P(F_{2,520} > 6.61) \approx 0.0015$. This is the probability of observing an F -statistic as large as 6.61 or larger, *if H_0 were true*. Since $0.0015 < 0.05 = \alpha$, we reject H_0 . In fact, we would reject at any significance level above 0.15% — the evidence is very strong.

Interpretation in words:

“At the 5% significance level, we reject the null hypothesis that previous experience has no effect on log-wages. The coefficients on $PrevExp$ and $PrevExp^2$ are *jointly* statistically significant, after controlling for current experience and education. The p -value of 0.0015 means that, if previous experience truly had no effect ($\beta_3 = \beta_4 = 0$), there would be only a 0.15% chance of observing an F -statistic as large as 6.61 purely due to sampling variation.”

Why “jointly” matters — a crucial nuance:

It is entirely possible that the individual t -tests for $\beta_3 = 0$ and $\beta_4 = 0$ fail to reject *either* coefficient separately. This can happen when $PrevExp$ and $PrevExp^2$ are highly correlated with each other (multicollinearity), which inflates their individual standard errors and makes each t -statistic small. But the F -test looks at their *combined* contribution. Even if neither coefficient is individually significant, together they may explain a significant amount of variation in wages. The F -test captures this joint effect that individual t -tests miss. This is precisely the scenario that motivates the F -test in the first place.

Step 6. R^2 version. When both models share the same dependent variable ($SST_r = SST_{ur} = SST$):

(g) Using $SSR = SST(1 - R^2)$, show that $F = \frac{(R_{ur}^2 - R_r^2)/q}{(1 - R_{ur}^2)/(n - k - 1)}$. Verify with $R_{ur}^2 = 0.3461$, $R_r^2 = 0.3294$.

Solution

Derivation, step by step.

Step 1: Express SSR in terms of R^2 . Recall the definition of R^2 :

$$R^2 = 1 - \frac{SSR}{SST} \quad \implies \quad SSR = SST(1 - R^2).$$

This gives us expressions for both models:

$$SSR_r = SST(1 - R_r^2), \quad SSR_{ur} = SST(1 - R_{ur}^2).$$

Crucially, both models have the same dependent variable ($\ln(Wage)$), so they share the same total sum of squares SST . This is what makes the substitution work.

Step 2: Substitute into the F -formula.

$$F = \frac{(SSR_r - SSR_{ur})/q}{SSR_{ur}/(n - k - 1)}$$

$$= \frac{[SST(1 - R_r^2) - SST(1 - R_{ur}^2)]/q}{SST(1 - R_{ur}^2)/(n - k - 1)}.$$

Step 3: Factor out SST . Since SST appears in every term, factor it out:

$$F = \frac{SST[(1 - R_r^2) - (1 - R_{ur}^2)]/q}{SST(1 - R_{ur}^2)/(n - k - 1)}.$$

The SST in the numerator and the SST in the denominator cancel:

$$F = \frac{[(1 - R_r^2) - (1 - R_{ur}^2)]/q}{(1 - R_{ur}^2)/(n - k - 1)}.$$

Step 4: Simplify the numerator.

$$(1 - R_r^2) - (1 - R_{ur}^2) = 1 - R_r^2 - 1 + R_{ur}^2 = R_{ur}^2 - R_r^2.$$

Therefore:

$$F = \frac{(R_{ur}^2 - R_r^2)/q}{(1 - R_{ur}^2)/(n - k - 1)}$$

Verification with Lecture 14 data.

Numerator:

$$R_{ur}^2 - R_r^2 = 0.3461 - 0.3294 = 0.0167.$$

This tells us that adding $PrevExp$ and $PrevExp^2$ increased the R^2 by 0.0167 (about 1.7 percentage points). Dividing by $q = 2$:

$$\frac{R_{ur}^2 - R_r^2}{q} = \frac{0.0167}{2} = 0.00835.$$

Denominator:

$$1 - R_{ur}^2 = 1 - 0.3461 = 0.6539.$$

This is the fraction of variation in $\ln(Wage)$ that the unrestricted model does *not* explain. Dividing by $n - k - 1 = 520$:

$$\frac{1 - R_{ur}^2}{n - k - 1} = \frac{0.6539}{520} = 0.001258.$$

F -statistic:

$$F = \frac{0.00835}{0.001258} \approx 6.64.$$

(The small difference from 6.61 computed in part (d) is due to rounding R^2 to four decimal places. With exact values, both formulas give identical results.)

Why this version is useful:

The R^2 formula makes the F -statistic particularly intuitive:

- **Numerator:** How much does R^2 *drop per restriction* when you impose H_0 ? If $R_{ur}^2 \approx R_r^2$ (the excluded variables barely increase R^2), the numerator is close to zero and F is small.
- **Denominator:** How much unexplained variation is there per degree of freedom in the unrestricted model? This is the “noise floor.”
- **Ratio:** Is the R^2 improvement from including $PrevExp$ and $PrevExp^2$ large relative to the noise? If so, these variables genuinely help explain wages and we should reject H_0 .